Questions: Arithmetic on algebraic fractions

Donald Campbell

Summary

A selection of questions for the study guide on arithmetic on algebraic fractions.

Before attempting these questions, it is highly recommended that you read Guide: Arithmetic on algebraic fractions.

Q1

Calculate the following additions and subtractions by first finding a common denominator. You should give your answer in its simplest form. You may assume that that all denominators are non-zero, so you do not need to write restrictions.

1.1. $\dfrac{2}{x}+\dfrac{5}{x}$

1.2. $\dfrac{x+1}{3}+\dfrac{x-2}{3}$

1.3. $\dfrac{4}{y}-\dfrac{1}{y}$

1.4. $\dfrac{3}{4x}+\dfrac{1}{4x}$

1.5. $\dfrac{b-3}{5}-\dfrac{b+2}{5}$

1.6. $\dfrac{1}{x}+\dfrac{1}{2x}$

1.7. $\dfrac{3}{5y}-\dfrac{2}{15y}$

1.8. $\dfrac{x}{3}+\dfrac{x}{6}$

1.9. $\dfrac{2}{x}+\dfrac{3}{x+1}$

1.10. $\dfrac{t+4}{t-1}-\dfrac{t-2}{t-1}$

1.11. $\dfrac{5}{x-3}+\dfrac{2}{2x-6}$

1.12. $\dfrac{2}{x+2} - \dfrac{1}{x-2}$

1.13. $\dfrac{c+1}{c^2} + \dfrac{2}{c}$

1.14. $\dfrac{3x}{x^2-9} + \dfrac{2x}{x+3}$

1.15. $\dfrac{x-1}{x+2} - \dfrac{2x+3}{x^2+4x+4}$

Q2

Calculate the following multiplications. Write your answer in its simplest form. Assume that all denominators are non-zero.

2.1. $\dfrac{x}{3} \cdot \dfrac{2}{5}$

2.2. $\dfrac{3a}{4} \cdot \dfrac{8}{a}$

2.3. $\dfrac{x+1}{2} \cdot \dfrac{x}{3}$

2.4. $\dfrac{5y}{2x} \cdot \dfrac{3x}{10y}$

2.5. $\dfrac{x-3}{x} \cdot \dfrac{x}{4}$

2.6. $-\dfrac{2x}{5} \cdot \dfrac{15}{x^2}$

2.7. $\dfrac{m^2-9}{m+3} \cdot \dfrac{m}{2m-6}$

2.8. $\dfrac{x+2}{x-1} \cdot \dfrac{x-1}{x+3}$

2.9. $\dfrac{3x}{x^2+2x} \cdot \dfrac{x+2}{4}$

2.10. $\dfrac{x^2+5x+6}{x+1} \cdot \dfrac{1}{x+2}$

2.11. $-\dfrac{z-4}{2z} \cdot \dfrac{3z}{z-4}$

2.12. $\dfrac{2x}{x^2-4} \cdot \dfrac{x+2}{3}$

Q3

Calculate the following divisions. Write your answer in its simplest form. Assume that all denominators are non-zero.

3.1. $\frac{t}{2} \div \frac{t}{5}$

3.2. $\frac{3}{x} \div \frac{1}{2x}$

3.3. $\frac{x+1}{4} \div \frac{x+1}{2}$

3.4. $\frac{2x}{3y} \div \frac{4}{9y}$

3.5. $\frac{x-2}{x} \div \frac{3}{4}$

3.6. $\frac{v^2-1}{v+1} \div v$

3.7. $-\frac{3x}{5} \div \frac{x}{10}$

3.8. $\frac{x^2+3x}{x} \div \frac{x+3}{2}$

3.9. $\frac{x}{x-4} \div \frac{2x}{x-4}$

3.10. $\frac{x^2-9}{x^2-3x} \div \frac{x-3}{x}$

3.11. $\frac{2n}{n+2} \div \frac{n}{n+2}$

3.12. $\frac{x^2-4}{2x} \div \frac{x-2}{3}$

Q4

Simplify the following compound algebraic fractions. Write your answer in its simplest form. Assume that all denominators are non-zero.

4.1. $\dfrac{\frac{1}{x}+\frac{1}{y}}{\frac{1}{x}}$

4.2. $\dfrac{-\frac{2x}{3}}{\frac{x+2}{9}}$

4.3. $\dfrac{\frac{x}{x-1}}{\frac{2x}{x-1}}$

4.4. $\dfrac{\dfrac{1}{t}+\dfrac{2}{t^2}}{\dfrac{3}{t}}$

4.5. $\dfrac{\dfrac{x+1}{2x}}{-\dfrac{x-1}{4x^2}}$

After attempting the questions above, please click this link to find the answers.

Version history and licensing

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